EXAM SHIELD

Computer Architecture and Operating Systems Question Bank

Processors, memory hierarchy, processes, scheduling and concurrency.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 21 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Computer Architecture and Operating Systems

Processors, memory hierarchy, processes, scheduling and concurrency.

1. Which CPU component performs arithmetic and logical operations?

A. Control unit
B. ALU
C. Program counter
D. Cache

Answer: B

2. Which of these memories is fastest?

A. Solid-state drive
B. Main memory (RAM)
C. L3 cache
D. CPU registers

Answer: D

3. Deadlock needs mutual exclusion, hold and wait, no preemption and:

A. Starvation
B. Circular wait
C. Paging
D. Thrashing

Answer: B

4. Which scheduling algorithm can starve long jobs?

A. Round robin
B. Fair-share scheduling
C. First-come, first-served
D. Shortest job first

Answer: D

5. Virtual memory lets a process:

A. Run without a CPU
B. Use more memory than is physically installed
C. Bypass the operating system
D. Share CPU registers with other processes

Answer: B

6. Code running in kernel mode differs from user mode because it:

A. Has unrestricted access to the hardware
B. Cannot handle interrupts
C. Has no memory of its own
D. Can only run applications

Answer: A

7. With byte addressing, a 32-bit address can reach at most:

A. 16 GB
B. 2 GB
C. 64 KB
D. 4 GB

Answer: D

8. Which number system uses base 16?

A. Decimal
B. Hexadecimal
C. Octal
D. Binary

Answer: B

9. The hexadecimal number FF equals _____ in decimal. [1 mark]

Answer: 255

10. When the CPU stops one process and starts another, the operating system performs a context _____. [1 mark]

Answer: switch

11. Excessive paging that keeps the CPU waiting for the disk is called _____. [1 mark]

Answer: thrashing

12. Explain the fetch-decode-execute cycle and how pipelining improves throughput. [5 marks]

13. Compare processes and threads, and explain how a race condition occurs and how it can be prevented. [5 marks]